Lab 2: Containerization & Kubernetes Orchestration

Task 2: Manage Docker Containers and Images

1. Container Lifecycle Management (list, start, stop, remove)

```
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker ps -a
CONTAINER ID   IMAGE              COMMAND           CREATED         STATUS         PORTS                                       NAMES
a590265f494c   my-flask-app:1.0   "python app.py"   9 minutes ago   Up 9 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app> docker stop flask-container
flask-container
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker ps -a
CONTAINER ID   IMAGE              COMMAND           CREATED          STATUS                      PORTS     NAMES
a590265f494c   my-flask-app:1.0   "python app.py"   10 minutes ago   Exited (137) 8 seconds ago            flask-container
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker start flask-container
flask-container
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker rm -f flask-container
flask-container
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>
```

2. Docker Image Tagging & Registry Operations

```
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker images

                                  i Info →  U  In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
devuser/my-flask-app:v1.0      5f87b4ab24f9       214MB           52MB
docker-lab-app-web:latest      42ca1352f1a9       214MB           52MB
docker-lab-app:1.0             5ee471abc4af       214MB           52MB
my-flask-app:1.0               5f87b4ab24f9       214MB           52MB
redis:alpine                   becdda6c7f4b       160MB         39.5MB
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker tag my-flask-app:1.0 devuser/my-flask-app:v1.0
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker tag my-flask-app:1.0 devuser/my-flask-app:latest
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker images

                                  i Info →  U  In Use
IMAGE                          ID             DISK USAGE   CONTENT SIZE   EXTRA
devuser/my-flask-app:latest    5f87b4ab24f9       214MB           52MB
devuser/my-flask-app:v1.0      5f87b4ab24f9       214MB           52MB
docker-lab-app-web:latest      42ca1352f1a9       214MB           52MB
docker-lab-app:1.0             5ee471abc4af       214MB           52MB
my-flask-app:1.0               5f87b4ab24f9       214MB           52MB
redis:alpine                   becdda6c7f4b       160MB         39.5MB
```

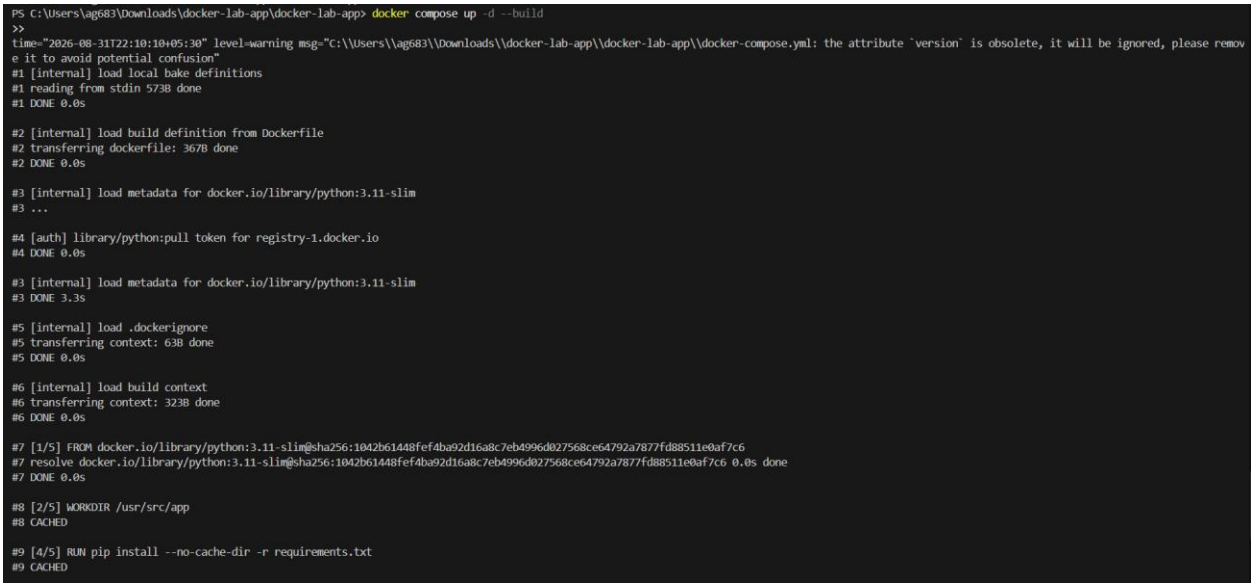
```
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app> docker push aditya5005/my-flask-app:v1.0
The push refers to repository [docker.io/aditya5005/my-flask-app]
44136fa355b3: Pushed
87e1b7cce023: Pushed
6310eb16bf42: Pushed
98ee5fd5bde3: Pushed
c86306e32cd0: Pushed
a14578096eda: Pushed
1cc6c39cb3f7: Pushed
f2af16e161b4: Pushed
23e0af4b0484: Pushed
22eea65fe735: Pushed
v1.0: digest: sha256:5f87b4ab24f9117b7594a418e869bc0e14fec7f216c3c55c094bab770eb09759 size: 856
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app> docker rmi devuser/my-flask-app:latest
Error response from daemon: No such image: devuser/my-flask
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app> docker rmi devuser/my-flask-app:v1.0
>>
Untagged: devuser/my-flask-app:v1.0
```

3. Multi-Container Setup with Docker Compose

3.1 Docker Compose Manifest

```yaml
version: '3.8'
▷Run All Services
services:
  ▷Run Service
  web:
    build: .
    ports:
      - "5000:5000"
    environment:
      - REDIS_HOST=redis
    depends_on:
      - redis
  ▷Run Service
  redis:
    image: redis:alpine
    ports:
      - "6379:6379"
```

3.2 Compose Execution & Dynamic Counter Verification



```
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app> docker compose up -d --build
>>
time="2026-08-31T22:10:10+05:30" level=warning msg="C:\\Users\\ag683\\Downloads\\docker-lab-app\\docker-lab-app\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remov
e it to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 573B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 367B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.11-slim
#3 ...

#4 [auth] library/python:pull token for registry-1.docker.io
#4 DONE 0.0s

#3 [internal] load metadata for docker.io/library/python:3.11-slim
#3 DONE 3.3s

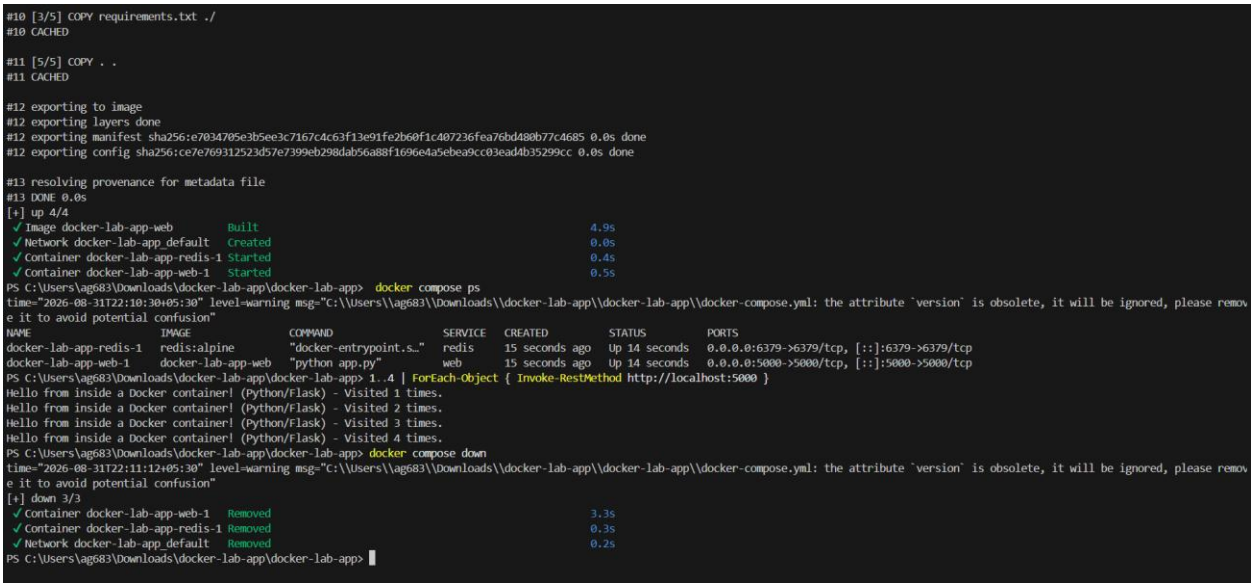
#5 [internal] load .dockerignore
#5 transferring context: 63B done
#5 DONE 0.0s

#6 [internal] load build context
#6 transferring context: 323B done
#6 DONE 0.0s

#7 [1/5] FROM docker.io/library/python:3.11-slim@sha256:1042b61448fef4ba92d16a8c7eb4996d027568ce64792a7877fd88511e0af7c6
#7 resolve docker.io/library/python:3.11-slim@sha256:1042b61448fef4ba92d16a8c7eb4996d027568ce64792a7877fd88511e0af7c6 0.0s done
#7 DONE 0.0s

#8 [2/5] WORKDIR /usr/src/app
#8 CACHED

#9 [4/5] RUN pip install --no-cache-dir -r requirements.txt
#9 CACHED
```



```
#10 [3/5] COPY requirements.txt ./
#10 CACHED

#11 [5/5] COPY . .
#11 CACHED

#12 exporting to image
#12 exporting layers done
#12 exporting manifest sha256:e7034705e3b5ee3c7167c4c63f13e91fe2b60f1c407236fea76bd480b77c4685 0.0s done
#12 exporting config sha256:ce7e769312523d57e7399eb298dab56a88f1696e4a5ebea9cc03ead4b35299cc 0.0s done

#13 resolving provenance for metadata file
#13 DONE 0.0s
[+] up 4/4
 ✓ Image docker-lab-app-web         Built                                    4.9s
 ✓ Network docker-lab-app_default   Created                                  0.0s
 ✓ Container docker-lab-app-redis-1 Started                                  0.4s
 ✓ Container docker-lab-app-web-1   Started                                  0.5s
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>  docker compose ps
time="2026-08-31T22:10:30+05:30" level=warning msg="C:\\Users\\ag683\\Downloads\\docker-lab-app\\docker-lab-app\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remov
e it to avoid potential confusion"
NAME                    IMAGE                COMMAND                  SERVICE   CREATED          STATUS          PORTS
docker-lab-app-redis-1  redis:alpine         "docker-entrypoint.s…"   redis     15 seconds ago   Up 14 seconds   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
docker-lab-app-web-1    docker-lab-app-web   "python app.py"          web       15 seconds ago   Up 14 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app> 1..4 | ForEach-Object { Invoke-RestMethod http://localhost:5000 }
Hello from inside a Docker container! (Python/Flask) - Visited 1 times.
Hello from inside a Docker container! (Python/Flask) - Visited 2 times.
Hello from inside a Docker container! (Python/Flask) - Visited 3 times.
Hello from inside a Docker container! (Python/Flask) - Visited 4 times.
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app> docker compose down
time="2026-08-31T22:11:12+05:30" level=warning msg="C:\\Users\\ag683\\Downloads\\docker-lab-app\\docker-lab-app\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remov
e it to avoid potential confusion"
[+] down 3/3
 ✓ Container docker-lab-app-web-1   Removed                                  3.3s
 ✓ Container docker-lab-app-redis-1 Removed                                  0.3s
 ✓ Network docker-lab-app_default   Removed                                  0.2s
PS C:\Users\ag683\Downloads\docker-lab-app\docker-lab-app>
```